1. Clinical Question & Core Hypothesis

Clinical Question

Tumors with dense, organized immune infiltrates (TLS/Immunoscore) often predict better outcomes, yet many TLS-rich MIBC tumors fail to respond to immunotherapy — why does local immune organization sometimes not translate into effective anti-tumor immunity? 2) Tier 1 paths used: (a) TGFB1/TGF-β signaling (core driver of fibrosis, EMT, and immune exclusion) provides a mechanistic axis that can physically and functionally prevent immune cells from engaging tumor cells; (b) Immunoscore/TLS-like lymphoid aggregates capture the imaging/immune organization whose effectiveness is in question; (c) Urine DNA methylation markers represent tumor-side epigenetic programs (e.g., antigen-presentation or EMT-associated methylation) that could silence immune-recognition. I built the hypothesis by linking TGFB1-mediated stromal/immune exclusion with epigenetic silencing signatures that together blunt TLS function. 3) Tier 2 evidence supporting the claim: PTGDS and HDL-C point to local inflammatory/prostaglandin and systemic lipid–immune modulation that can amplify or attenuate TGF-β effects and influence immune cell function; lymph node metastasis as a clinical correlate reinforces that an immune-excluded, fibrotic phenotype is clinically aggressive. 4) Why therapy_response chosen as the endpoint: The key translational question is not only prognosis but whether TLS presence predicts benefit from immune checkpoint blockade; TGF-β and methylation mechanisms are well-positioned to determine immunotherapy responsiveness, making therapy response the most relevant primary endpoint. 5) What I excluded and why: I deliberately excluded Tier 3 exploratory links (PSA cross-disease signals, CSF/AD biomarker analogies, stool SDC2/TFPI2 markers) because they are indirect to bladder-specific immune–fibrosis–epigenetic biology and would diffuse the mechanistic focus?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), TLS-like lymphoid aggregates indicate potential local anti-tumor immunity, but high TGFB1 activity drives fibrotic stromal remodeling and immune exclusion, compromising TLS maturation and function. Concurrent tumor DNA hypermethylation silences antigen presentation programs, further diminishing TLS-driven immune engagement. As a result, TLS-rich tumors with elevated TGFB1 activity and high urine DNA methylation scores will exhibit reduced response to immune checkpoint blockade compared with TLS-rich tumors that have low TGFB1 activity and low methylation; this effect will be detectable as a statistical interaction between TGFB1 activity and urine DNA methylation on therapy_response, independent of TLS presence.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TGFB1, Urine DNA methylation markers
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1/TGF-β signaling (core driver of fibrosis, EMT, and immune exclusion) provides a mechanistic axis that can physically and functionally prevent immune cells from engaging tumor cells; (b) Immunoscore/TLS-like lymphoid aggregates capture the imaging/immune organization whose effectiveness is in question; (c) Urine DNA methylation markers represent tumor-side epigenetic programs (e.

Urine DNA methylation markers: Urine DNA methylation markers represent tumor-side epigenetic programs (e.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	Lymph node metastasis	Supporting modifier from knowledge graph
Secondary	Immunoscore (CD3/CD8 densities)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.